DAY 4

From Concepts to Proof of Concept

Product design approach and PoC realization

SysML v2 / Real components / Agentic design

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THE POC REALIZATION JOURNEY

01 **IMPLEMENT**
Assemble a PoC from available technology

02 **ASSESS**
Test feasibility against explicit criteria

03 **INNOVATE**
Address needs current options cannot meet

DAY 4 / CLASS SCHEDULE

3 H 15 MIN

Day 4: Product design approach and PoC realization

Build from a product concept toward an evidence-backed proof of concept.

4.1 Product design approach and class discussion — 90 MIN

- Proofs of concepts and primitives — 60 MIN

  Lead designer and constructive pressure
  Concept versus PoC and Idea Wall model examples
  PoC realization, innovation, and agent skills

- Class discussion: student concepts — 30 MIN — Each student shares the concept in a 2–3 minute presentation.

BREAK — 15 MIN

4.2 Practice: proofs of concept in SysML v2 — 90 MIN

- Model structure, behavior, and analysis. Trace PoC elements to requirements, design decisions, and test evidence.

SECTION 4.1.1 / DESIGN LEADERSHIP

2 MIN

The student is the lead designer

The agent expands the design space; the student remains accountable for the direction.

DESIGN ANCHOR

The user

Needs, context, constraints, and the experience the product should make possible.

›

DESIGN AUTHORITY

The student

Sets intent and criteria, weighs trade-offs, judges evidence, and owns the decision.

›

DESIGN SUPPORT

The agent

Finds candidates, drafts model elements, compares evidence, and raises questions.

CORE RESPONSIBILITY

The student can explain why this realization serves the intended user.

SECTION 4.1.1 / DESIGN LEADERSHIP

4 MIN

The concept describes intent; the PoC tests a realization

Keep the user outcome in view while exploring concrete ways to achieve it.

CONCEPT MODEL

What should be true for the user?

Describes the intended value and behavior.
Keeps implementation choices open.
Makes assumptions and criteria explicit.

›

POC REALIZATION

Can this route deliver it?

Assembles available parts and software.
Tests one criterion; exposes manual steps.
A missing capability can become new design.

DESIGN SPACE

One concept can produce multiple PoCs, each with its own question and evidence.

SECTION 4.1.1 / DESIGN LEADERSHIP

4 MIN

Idea Wall: trace one criterion into a real assembly

Android voice input and Google Keep form one testable route; the model keeps its limits visible.

SYSML V2 / TEACHING MODEL

```
package IdeaWallPoC {
  abstract part def IdeaWallSystem;
  part def AndroidVoiceToText;
  part def GoogleKeep;
  requirement def CaptureOnTheGo {
    subject system : IdeaWallSystem;
    doc /* Capture ideas on the move without typing. */
  }
  requirement captureOnTheGo : CaptureOnTheGo;
  part def IdeaWallRealization :> IdeaWallSystem {
    part voiceInput : AndroidVoiceToText;
    part ideaStore : GoogleKeep;
    // Google Takeout export remains manual.
  }
  part wallPoC : IdeaWallRealization;
  satisfy captureOnTheGo by wallPoC;
}
```

REAL PARTS

Android voice-to-text + Google Keep

VISIBLE LIMITS

Takeout export is manual; browser zoom simulates zooming ideas.

WHAT THE TRACE SAYS

The PoC is intended to address the criterion. A user test supplies evidence.

SECTION 4.1.1 / DESIGN LEADERSHIP

2 MIN

A PoC is a learning experiment

Realize enough of the concept to learn, while making incomplete parts explicit.

01

Question

Which assumption matters most to the user?

›

02

Realize

Build only what the question needs.

›

03

Observe

Record behavior, outcomes, and gaps.

›

04

Refine

Update the concept and next test.

POC DISCIPLINE **A manual or simulated step is acceptable when its limits are visible.**

SECTION 4.1.1 / DESIGN LEADERSHIP

3 MIN

Trace the design rationale

A testable claim should connect its origin to a realization and observable evidence.

ORIGIN
Assumption
Ideas can be captured while moving.

›

TEST
Criterion
Capture an idea without typing.

›

CONCEPT
Behavior
Voice capture of an idea.

›

REALIZATION
PoC parts
Android voice input + Keep.

›

OBSERVATION
Evidence
Run a user test; record result.

SYSML V2 TRACE **satisfy captureWithoutTyping by voiceInput;**

"by" names the feature that claims to satisfy the requirement.

READ THE PATH **No link means an open question to review, not an automatic verdict.**

SECTION 4.1.1 / DESIGN LEADERSHIP

2 MIN

Trade-offs are documented design decisions

Record why this realization fits the test and which limits remain open.

REQUIREMENT PATH

What are we testing?

Voice capture should let a participant record an idea without typing.

›

DECISION RECORD

Why this realization?

Reuse an existing app to test the capture experience quickly.

›

ACCEPTED LIMIT

What remains open?

The PoC does not yet establish continuous analytics or production integration.

Decision note = choice + rationale + accepted limitation

SECTION 4.1.1 / DESIGN LEADERSHIP

3 MIN

Constructive pressure stress-tests design maturity

Challenge the design rationale so hidden assumptions become visible and testable.

01	USER	Whose situation or access needs are missing?
02	ASSUMPTION	Which component claim is unverified?
03	FAILURE	What condition could make this realization fail?
04	EVIDENCE	What observation could weaken the claim?

MATURITY SIGNAL — **A challenge leads to a test, a revision, or a justified decision.**

SECTION 4.2.1 / CONCEPT TO REALIZATION

2 MIN

The concept sets intent; the PoC tests one route

Keep the user outcome stable while making one implementation concrete and testable.

CONCEPT MODEL

What should be true for the user?

User value and intended behavior
Technology-agnostic
Criteria and assumptions made explicit

›

PROOF OF CONCEPT

Can this route deliver it?

One selected realization
Real parts, software, and operations
Evidence for a bounded question

Concept = intended user value. | PoC = one evidence-producing realization.

SECTION 4.2.1 / CONCEPT TO REALIZATION

3 MIN

Idea Wall: trace one criterion into an assembly

One article criterion becomes one concrete realization with its limits visible.

SYSML V2 / TEACHING EXCERPT

```
package IdeaWallPoC {
  abstract part def IdeaWallSystem;
  part def AndroidVoiceToText;
  part def GoogleKeep;
  requirement def CaptureOnTheGo {
    subject system : IdeaWallSystem;
    doc /* Capture ideas on the move without typing. */
  }
  requirement captureOnTheGo : CaptureOnTheGo;
  part def IdeaWallRealization :> IdeaWallSystem {
    part voiceInput : AndroidVoiceToText;
    part ideaStore : GoogleKeep;
  }
  part wallPoC : IdeaWallRealization;
  satisfy captureOnTheGo by wallPoC;
}
```

SELECTED TECHNOLOGY

Android voice input + Google Keep

KNOWN LIMITS

Export is manual; zoom behavior is simulated.

TRACE ≠ TEST RESULT

The model records intent. A user test supplies evidence.

SECTION 4.2.1 / CONCEPT TO REALIZATION

2 MIN

One concept can branch into several PoCs

Keep the criterion constant; vary the realization to expose trade-offs and gaps.

BRANCH A / ARTICLE

Assemble what exists

Android voice input
Google Keep as idea wall
Test capture without typing

BRANCH B / BOUNDED

Accept a manual step

Use Takeout for export
Simulate display zoom
Test capture, not automation

BRANCH C / EXPLORE

Design the missing part

Shared dashboard is unmet
Specify collaboration need
Investigate new software

A missing capability can be the beginning of a new design.

SECTION 4.2.1 / CONCEPT TO REALIZATION

1 MIN

Start procurement with the criterion

Define what must be learned before asking an agent to find a component.

01 CRITERION	**Capture ideas on the move without typing.**
02 SPECIFY	**Mobile voice input; save and retrieve text.**
03 SEARCH + VERIFY	**Check vendor, app/API, and repository evidence.**
04 GAP → DESIGN	**If no fit, define the missing capability.**

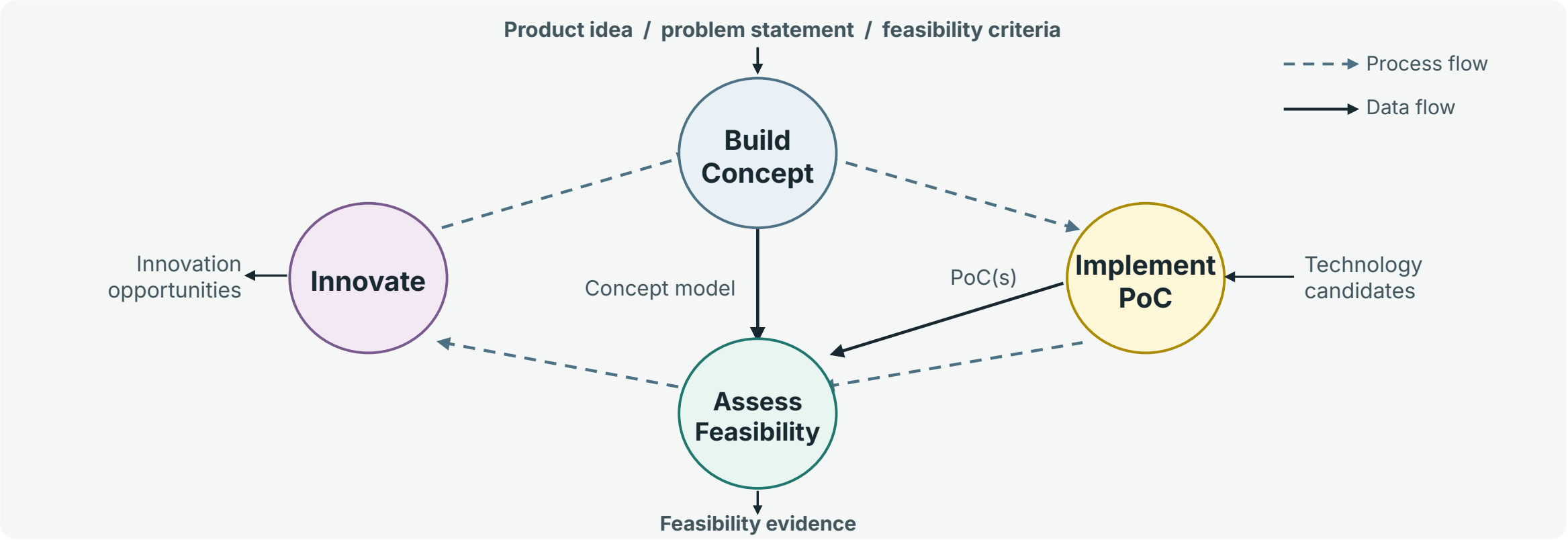
Next: use the component procurement skill to build a verified shortlist.

SECTION 4.2.1 / POC REALIZATION AND INNOVATION

4 MIN

Visual Concept Making connects the concept to evidence

The cycle begins with a user problem and criteria; evidence and new opportunities inform what comes next.



SECTION 4.2.1 / POC REALIZATION AND INNOVATION

4 MIN

Implement PoC by matching the model to real technology

Use the concept's functions, properties, and interfaces to search for candidate parts and software.

1 / CONCEPT SPECIFICATION

What must the part do?

Operation: capture an idea
Property: works on the go
Criterion: no typing required

›

2 / TECHNOLOGY CANDIDATES

What already exists?

Hardware, apps, APIs, services, code
Compare candidate behavior and interfaces against the specification

›

3 / POC REALIZATION

What can we assemble?

Select concrete parts and software
Integrate them into one or more PoCs
Mark manual or simulated steps

IDEA WALL / ARTICLE EXAMPLE

Voice capture without typing: Android voice-to-text + Google Keep as a concrete first PoC

The student chooses what the PoC must teach and how much of the experience can be simulated.

SECTION 4.2.1 / POC REALIZATION AND INNOVATION

4 MIN

Assess feasibility by tracing criteria to evidence

Compare the concept criterion with the PoC realization, then label what is demonstrated, simulated, or open.

CONCEPT CRITERION	2019 ARTICLE PoC	ASSESSMENT
Voice capture without typing	Android voice-to-text + Keep	Realized in the PoC; test the experience
Shared board for participants	One shared account workaround	Simulated collaboration; has a scale limit
Continuous analytics	Takeout export and manual processing	Manual step remains; automation is open

ASK OF EACH ROW

What evidence did we collect? What remains untested? What decision does this support?

Historical article example; the table summarizes the author's 2019 PoC and its stated limits.

SECTION 4.2.1 / POC REALIZATION AND INNOVATION

3 MIN

An unmet criterion can reveal an innovation opportunity

When candidates leave a gap, examine integration and decide whether a new part or software is needed.

THE GAP

A criterion has no match

The existing parts do not yet realize the required behavior or interface.

›

DESIGN RESPONSE

Try a new combination

Integrate candidates differently, adapt a part, or write the missing software.

›

MODEL THE OPPORTUNITY

Make the new work explicit

Add the capability and interfaces to the model; trace them to the criterion.

IDEA WALL / ARTICLE EXAMPLE

Manual Takeout export suggests an automation gap to investigate.

DAY 5 **Alternatives, innovation types, and iteration of the loop.**

Innovation can begin with a documented gap in the realization.

SECTION 4.2.2 / EXAMPLES OF POC MODELS

2 MIN

We will examine the Idea Wall PoC through four views

Each view answers a different question; traceability connects them to the criterion.

MODEL VIEW	WHAT IT MAKES EXPLICIT	IDEA WALL EXAMPLE
Structure	What is assembled, and how do parts connect?	Voice input → idea store → shared wall; mark real, manual, or simulated parts.
Behavior	What happens from capture to use?	Speak an idea → transcribe → save → display → export.
Analysis	What criterion are we testing, and how?	Test “capture without typing”; observe completion, transcription accuracy, and time.
Test & evidence	What result would support or weaken the claim?	Run a user task; record outcome and limitations, then trace evidence to the criterion.

CROSS-CUTTING **Requirement trace • design decision • limitation • test evidence**

SECTION 4.2.2 / EXAMPLES OF POC MODELS

3 MIN

Structure: available parts form the PoC assembly

The example uses Android voice capture and Google Keep, with a few limits left visible.

SYSML V2 / STRUCTURE EXCERPT

```
package IdeaWallPoC {
  part def AndroidPhone;
  part def AndroidVoiceToText;
  part def GoogleKeepMobile;
  part def ChromeBrowser;
  part def GoogleKeepWeb;
  part def Display4K;
  part def KeepCloudService;
  part def IdeaWallSystem;
  part def MobileVoiceClient {
    part device : AndroidPhone;
    part speech : AndroidVoiceToText;
    part notes : GoogleKeepMobile;
  }
  part def SharedWallClient {
    part browser : ChromeBrowser;
    part notes : GoogleKeepWeb;
    part screen : Display4K;
  }
  part def IdeaWallRealization :> IdeaWallSystem {
    doc /* Shared board uses one account; Takeout export is manual. */
    part mobile : MobileVoiceClient;
    part sharedWall : SharedWallClient;
    part storage : KeepCloudService;
  }
  part wallPoC : IdeaWallRealization;
}
```

PARTS IN THE EXCERPT

Mobile / Android phone, voice input, Keep
Shared wall / Chrome, Keep Web, 4K display
Storage / Keep cloud service

LIMITS THAT REMAIN

Same-account workaround; random logouts began at around 20 participants. Takeout stayed manual.

READ THE MODEL

:> means specializes; nested part usages show this realization's composition.

SECTION 4.2.2 / EXAMPLES OF POC MODELS

3 MIN

Behavior: one spoken idea becomes a shared note

The model orders transcription, saving, synchronization, and display, while tracing the idea text through each step.

SYSML V2 / BEHAVIOR EXCERPT

```
package IdeaWallPoC {
  item def Speech;
  item def IdeaText;
  action def CaptureAndDisplayIdea {
    in item spokenIdea : Speech;
    action transcribe {
      in item audio : Speech;
      out item text : IdeaText;
    }
    action saveToKeep {
      in item text : IdeaText;
      out item savedNote : IdeaText;
    }
    action syncToWall {
      in item savedNote : IdeaText;
      out item sharedNote : IdeaText;
    }
    action displayOnWall {
      in item sharedNote : IdeaText;
    }
    first transcribe then saveToKeep;
    first saveToKeep then syncToWall;
    first syncToWall then displayOnWall;
    flow spokenIdea to transcribe.audio;
    flow transcribe.text to saveToKeep.text;
    flow saveToKeep.savedNote to syncToWall.savedNote;
    flow syncToWall.sharedNote to displayOnWall.sharedNote;
  }
}
```

WHAT HAPPENS

1 Android voice-to-text turns speech into text
2 Keep saves the text
3 Keep syncs it to the shared wall
4 Chrome displays the note

READ THE NOTATION

first / then sets action order. flow connects the data passed between actions.

WHAT THIS DOES NOT PROVE

One-account sharing is a workaround; this sequence does not establish robust multi-user collaboration.

SECTION 4.2.2 / EXAMPLES OF POC MODELS

3 MIN

Analysis makes success measurable

Define the evidence and acceptance targets before testing; the model does not invent results.

SYSML V2 / TEXTUAL MODEL

```
package IdeaWallPoC {
  import ScalarValues::*;
  requirement def CaptureWithoutTyping {
    subject system : IdeaWallSystem;
    doc /* Capture an idea on the go without typing. */
  }
  item def CaptureMeasures {
    attribute taskSuccessRate : Real;
    attribute transcriptionErrorRate : Real;
    attribute elapsedTimeSeconds : Real;
  }
  analysis def CaptureAnalysis {
    subject system : IdeaWallRealization;
    return measures : CaptureMeasures;
    objective {
      doc /* Assess voice capture against the criterion. */
      requirement : CaptureWithoutTyping;
    }
  }
}
```

MEASURES

Task success rate
Transcription error rate
Elapsed time per capture

DECIDE BEFORE TESTING

Set the task, speaking conditions, sample, and target thresholds with the user.

MODEL BOUNDARY

This case defines the question and result type. It contains no measured values or success claim.

SECTION 4.2.2 / EXAMPLES OF POC MODELS

3 MIN

Verification connects test evidence to the requirement

A verification case names the PoC, the criterion, the evidence path, and the verdict.

SYSML V2 / TEXTUAL MODEL

```
package IdeaWallPoC {
  import VerificationCases::*;
  part def Participant;
  verification def VoiceCaptureTest {
    subject system : IdeaWallRealization;
    actor participant : Participant;
    objective {
      doc /* Test capture by voice and availability on the wall. */
      verify CaptureWithoutTyping;
    }
    action runCaptureTask;
    action recordEvidence {
      out item measures : CaptureMeasures;
    }
    action judgeEvidence {
      in item measures : CaptureMeasures;
      out verdict : VerdictKind;
    }
    first runCaptureTask then recordEvidence;
    first recordEvidence then judgeEvidence;
    flow recordEvidence.measures to judgeEvidence.measures;
    return verdict : VerdictKind = judgeEvidence.verdict;
  }
}
```

TEST SCENARIO

One participant captures a spoken idea; the note should be available on the shared wall.

EVIDENCE TO RECORD

Capture completed, transcript compared with speech, elapsed time, and observed recovery issues.

VERDICT

Pass, fail, inconclusive, or error. Set acceptance thresholds before collecting evidence.

SECTION 4.3.1 / AGENT UPDATE

3 MIN

How to equip OpenCode for the PoC stage?

The agent needs repeatable skills for tracing the design, finding technologies, and modeling selected components.

CHECK THE MODEL

TRACEABILITY-AUDIT

Find missing links

Flag elements without an origin, requirements without a realization, criteria without evidence, and decisions without rationale.

Output: reviewable findings

SEARCH THE ECOSYSTEM

COMPONENT-PROCUREMENT

Find candidate technologies

Turn modeled functions, interfaces, constraints, and use cases into supplier or software searches with evidence and open questions.

Output: candidate comparison

MODEL A SELECTION

COMPONENT-TO-SYSML-V2

Represent the selected part

Translate documented component or API details into an approximate, reusable SysML v2 definition.

Output: facts + inferred choices

The agent supplies methods and evidence. The student decides what the evidence means and what becomes the design.

SECTION 4.3.1 / AGENT UPDATE

3 MIN

The Traceability-Audit skill finds model gaps

OpenCode checks explicit model links and returns findings for the student to judge.

ELEMENT WITHOUT ORIGIN

Which part has no trace to a requirement or assumption?

REQUIREMENT WITHOUT REALIZATION

Which requirement has no linked concept or PoC element?

CRITERION WITHOUT EVIDENCE

Which criterion has no linked test result or observation?

DECISION WITHOUT RATIONALE

Which realization choice lacks a documented reason?

SKILL OUTPUT Element ID / Missing link / Model evidence / Question for the student

The skill flags gaps. The lead designer interprets and resolves them.

SECTION 4.3.1 / AGENT UPDATE

3 MIN

The procurement skill searches from design intent

Searches begin with the model and return candidates with evidence, trade-offs, and open questions.

FROM THE CONCEPT MODEL

Idea Wall mobile capture

FUNCTION	Capture an idea
CRITERION	No typing while moving
INTERFACE	Mobile input → note
CONSTRAINT	Fast PoC; low integration effort

SEARCH ECOSYSTEMS

Focused queries

- Supplier catalogs
- Datasheets + API docs
- Software ecosystems
- Integration examples

Search terms inherit the model's criteria and constraints.

CANDIDATE REPORT

CANDIDATE	What product or software?
EVIDENCE	Which source supports the match?
TRADE-OFF	What requirement or constraint is weak?
STATUS	What availability or license is verified?
OPEN QUESTION	What needs supplier or user confirmation?

A search result is a candidate to evaluate, not a design decision.

SECTION 4.3.1 / AGENT UPDATE

3 MIN

The component-to-SysML skill preserves evidence

A selected datasheet or API becomes an approximate model, with facts kept separate from inference.

INPUT

Android SpeechRecognizer API

PROVENANCE CHECK

DOCUMENTED
RECORD_AUDIO permission; listener returns results.

LIMIT
May stream audio remotely; not for continuous use.

APPROXIMATE SYSML V2 EXCERPT

```
package AndroidSpeechCandidate {
  item def AudioInput;
  item def RecognitionResults;
  part def AndroidSpeechRecognizer {
    attribute requiresRecordAudioPermission : Boolean;
    action def recognize {
      in audio : AudioInput;
      out results : RecognitionResults;
    }
    doc /* Documented: permission + result listener.
           Inferred: recognize abstracts API lifecycle.
           Source: Android SpeechRecognizer API. */
  }
}
```

SOURCE-BACKED FACT

INFERRED ABSTRACTION

Student reviews before reuse in the PoC model.

SECTION 4.3.1 / AGENT UPDATE

2 MIN

Students can add their own OpenCode skills

A reusable SKILL.md packages a clear trigger and a repeatable workflow.

01 / CREATE THE FILE

Choose a project path

```
.opencode/skills/
<skill-name>/SKILL.md
```

Project-local skills travel with the repository.

02 / DEFINE THE WORKFLOW

Name it and say when to use it

```
---
name: my-poc-review
description: Review a PoC model
---
```

Add steps, expected output, and limits below.

03 / OPEN + TEST

Check discovery and refine

Open the project in OpenCode. Confirm the skill appears, try it on a real task, and revise the instructions.

Required frontmatter: name + description

Start with a task you repeat. Make the agent's trigger, steps, output, and limits explicit.

SECTION 4.1.1 / TRACEABILITY AUDIT

2 MIN

Practical example of Skill: traceability-audit

```
---
name: traceability-audit
description: Audit SysML v2 textual models for missing requirement-to-realization links, design elements without a stated requirement or assumption origin, unresolved trace references, and gaps between modeled satisfaction and verification
evidence.
compatibility: opencode
metadata:
  course: keio-2026-agentic-product-concept-modeling
  model-language: sysml-v2
  opencode/slash: "true"
---

# Traceability Audit

## Purpose

Review a SysML v2 textual model for traceability gaps that deserve the lead designer's attention. Find:

1. In-scope product design or realization elements with no explicit origin in a requirement or a named assumption.
2. Requirements with no modeled realization link.
3. Negated satisfaction claims and unresolved or ambiguous references.
4. Requirements whose modeled satisfaction is not accompanied by an identifiable verification case, test result, or other stated evidence, when the project expects that evidence in the model or workspace.

The audit is a review aid. It does not decide whether a design is good, whether a requirement is correct, or whether the product works. The student or project designer interprets every finding and owns any change.

## When to use this skill

Use this skill when asked to audit, review, or stress-test traceability in a SysML v2 model or when asked to find:

- design elements with no requirement or assumption origin;
- requirements with no design element realizing them;
- broken or unclear trace links;
- satisfaction claims without test or validation evidence.

If the user provides a specific model file or package, use that scope. If several unrelated models are present and the intended scope is not clear, ask which one to review. Otherwise, audit the relevant SysML v2 model files in the current
project.
```

COURSE / END

The student is the lead designer.

Models make intent visible.

Agents extend the design work. Students judge evidence.

Questions?

THREE COURSE PRINCIPLES

01 **CONCEPT**
Describe the user need and intended experience

02 **PROOF OF CONCEPT**
Test a realization against explicit criteria

03 **HUMAN JUDGMENT**
The student decides what evidence means